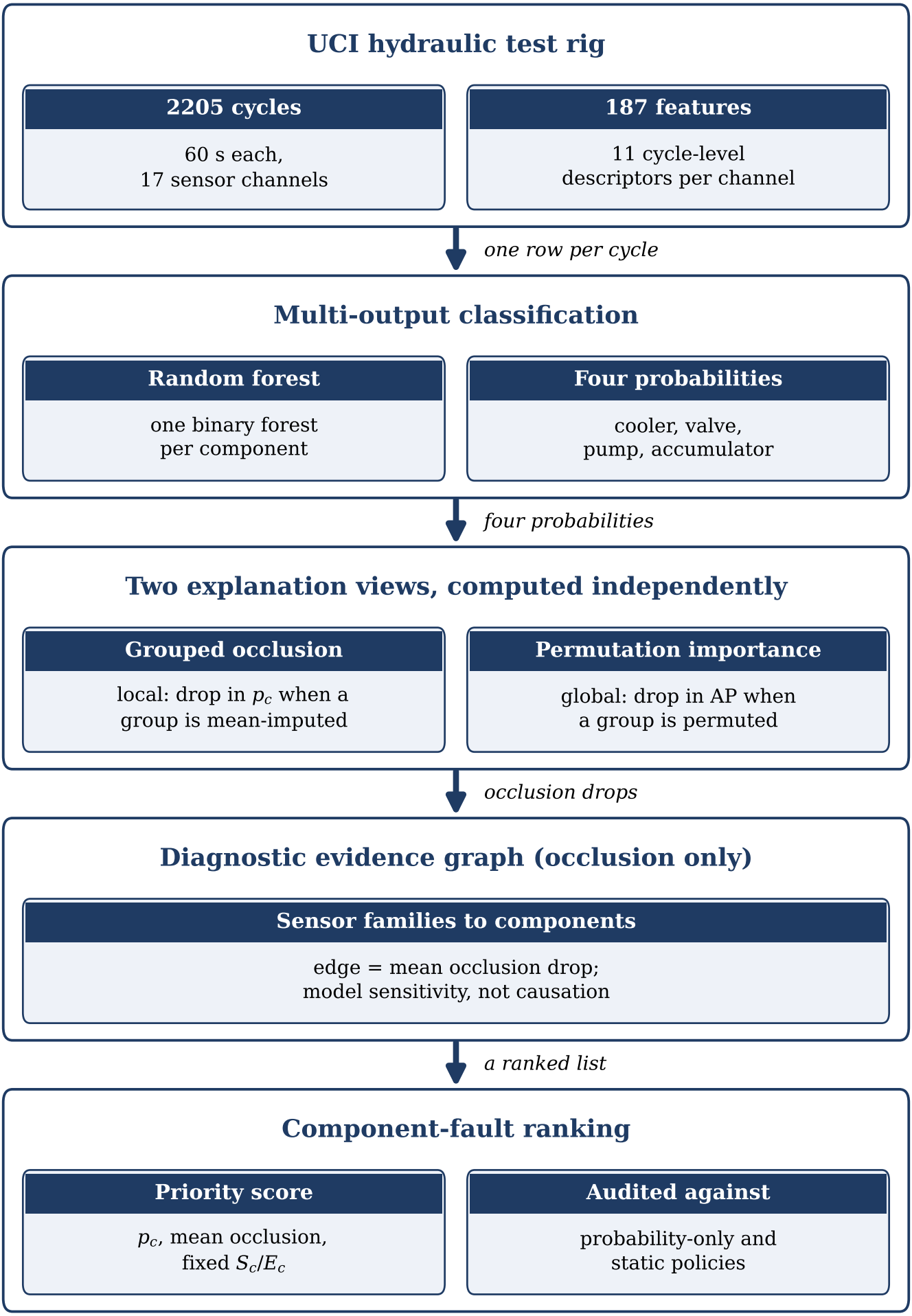
UCI hydraulic test rig
2205 cycles
60 s each,
17 sensor channels
187 features
11 cycle-level
descriptors per channel
one row per cycle
Multi-output classification
Random forest
one binary forest
per component
Four probabilities
cooler, valve,
pump, accumulator
four probabilities
Two explanation views, computed independently
Grouped occlusion
local: drop in p_c when a
group is mean-imputed
Permutation importance
global: drop in AP when
a group is permuted
occlusion drops
Diagnostic evidence graph (occlusion only)
Sensor families to components
edge = mean occlusion drop;
model sensitivity, not causation
a ranked list
Component-fault ranking
Priority score
p_c, mean occlusion,
fixed S_c/E_c
Audited against
probability-only and
static policies